

TRS-80 Color Computer 1/2 — Pinouts

6809E · 6847 VDG · 6821 PIA · 6883 SAM · MEMORY

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

MC6809E CPU

40-DIP

Vss	1	40	/HALT
/NMI	2	39	TSC
/IRQ	3	38	LIC
/FIRQ	4	37	/RESET
BS	5	36	AVMA
BA	6	35	Q
Vcc	7	34	E
A0	8	33	BUSY
A1	9	32	R/W
A2	10	31	D0
A3	11	30	D1
A4	12	29	D2
A5	13	28	D3
A6	14	27	D4
A7	15	26	D5
A8	16	25	D6
A9	17	24	D7
A10	18	23	A15
A11	19	22	A14
A12	20	21	A13

CoCo 1/2 CPU @ 0.89/1.79 MHz. The SAM drives its clocks (E,Q). **Fail:** dead machine, no boot.

MC6847 VDG video

40-DIP

Vss	1	40	DD7
DD6	2	39	CSS
DD0	3	38	HS
DD1	4	37	FS
DD2	5	36	RP
DD3	6	35	A/G
DD4	7	34	A/S
DD5	8	33	CLK
CHB	9	32	INV
φB	10	31	INT/EXT
φA	11	30	GM0
/MS	12	29	GM1
DA5	13	28	Y
DA6	14	27	GM2
DA7	15	26	DA4
DA8	16	25	DA3
Vcc	17	24	DA2
DA9	18	23	DA1
DA10	19	22	DA0
DA11	20	21	DA12

Video Display Generator. Pinout verified against the Motorola MC6847 datasheet. CoCo3 replaces the 6847+SAM with the GIME ASIC. **Fail:** no/bad video.

MC6821 PIA ×2

40-DIP

Vss	1	40	CA1
PA0	2	39	CA2
PA1	3	38	/IRQA
PA2	4	37	/IRQB
PA3	5	36	RS0
PA4	6	35	RS1
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	φ2/E
PB7	17	24	CS1
CB1	18	23	/CS2
CB2	19	22	CS0
Vcc	20	21	R/W

Two PIAs: U8 (\$FF00) keyboard/joystick/VDG-mode, U4 (\$FF20) sound/cassette/serial. MC6821 = MC6820 = 6520. **Fail:** dead keys, no sound/cassette.

MC6883 SAM sync addr mux

40-DIP

A11	1	40	Vcc
A10	2	39	A12
A9	3	38	A13
A8	4	37	A14
OSCin	5	36	A15
OSCout	6	35	MA7
VClk	7	34	MA6
DA0	8	33	MA5
/HS	9	32	MA4
/WE	10	31	MA3
/CAS	11	30	MA2
/RAS	12	29	MA1
Q	13	28	MA0
E	14	27	S0
R/W	15	26	S1
A0	16	25	S2
A1	17	24	A7
A2	18	23	A6
A3	19	22	A5
VSS	20	21	A4

Synchronous Address Multiplexer (SN74LS783): makes all clocks from a 14.31818 MHz xtal (OSCin/out), DRAM /RAS /CAS + MA0-MA7 row/col mux, and device selects S0-S2. Pinout verified against the CoCo service-manual schematic. **Fail:** dead machine, no video timing.

Memory & ROM

RAM/ROM

RAM	4116 (CoCo1) or 4164 (CoCo2), 16/32/64KB
ROM	Color BASIC 8/16KB (mask)
Cartridge	Program Pak ROM via slot
CoCo3	512KB + GIME ASIC (see CoCo3 sheet)
SAM controls RAM size strapping. CoCo3's GIME replaces SAM+VDG (separate sheet).	

Test & Repair

tips

No boot	6809E / SAM / ROM
No / bad video	6847 / SAM
Dead keys	U8 PIA
No sound/tape	U4 PIA
CoCo3 only	GIME ASIC
SAM faults look like total failure (it makes the clocks). VDG faults are video-only.	

Notes

△ verify these

RAM type	4116 vs 4164 by board rev
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(6809E, 6847, 6821, SAM verified)

Amber pins and **△ flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.